How a Question Is Answered

Steps 6–7 · `src/qa/`, `src/ui/views/ask.py` · `streamlit run src/ui/app.py` · 2026-10-28

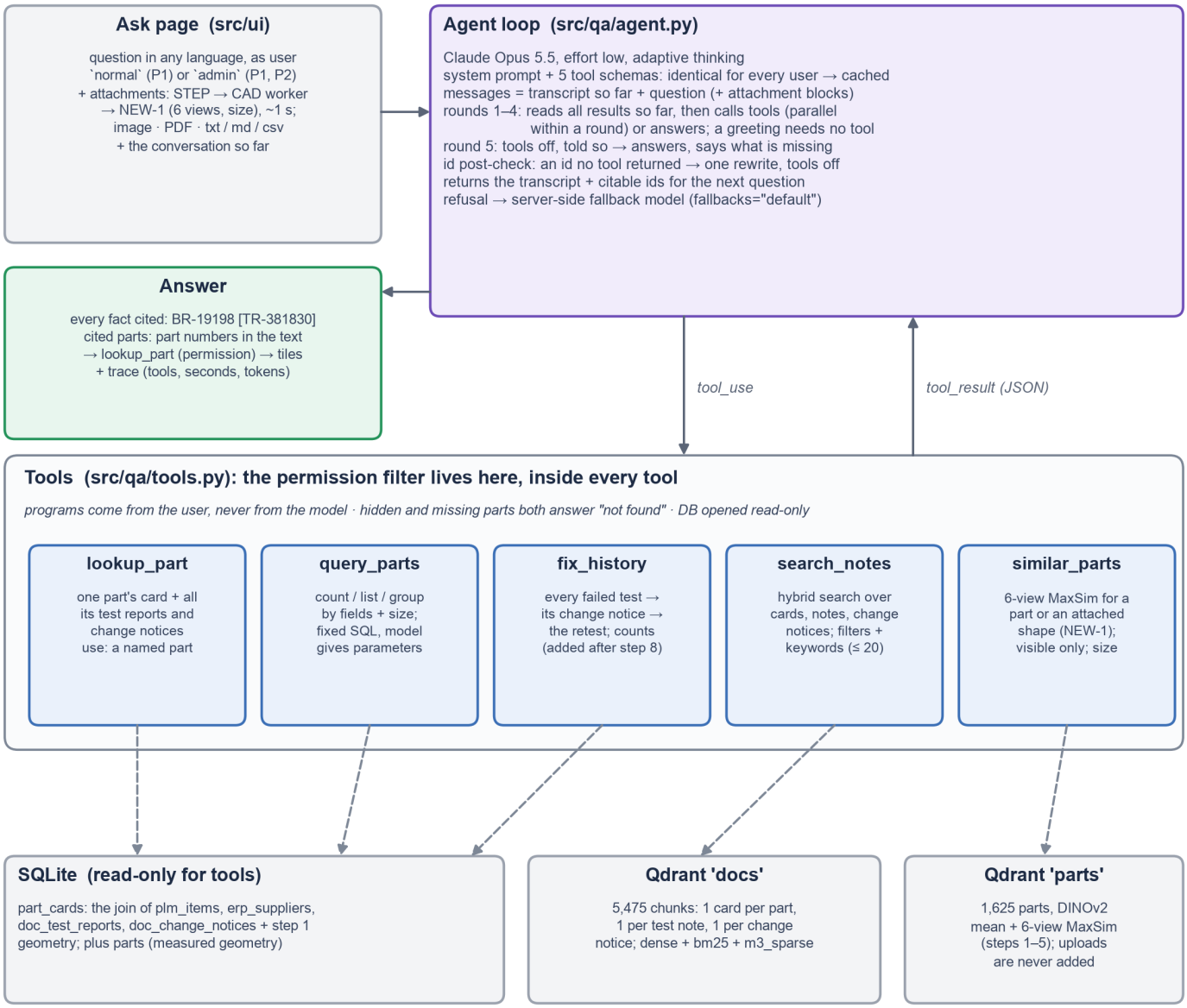


Figure 1. Claude chooses the tools; the tools decide what the user may see; the answer cites every fact.

Prompt rules

Hard facts go into filters (search can't see negation); bilingual synonyms as keywords; cite only ids a tool returned; name the revision a result belongs to; never guess. Attached files are context, not evidence; follow-ups may reuse earlier tool results.

Tools

Tool	Model controls	Permission
lookup_part	part number	hidden = missing → 'not found'
query_parts	parameters only; the SQL is fixed	program condition added by code
search_notes	query, keywords, exact filters	filter inside each prefetch
fix_history	test type, class, material ('any' = all)	program condition added by code
similar_parts	part number or attached shape (NEW-1), size ratio	only visible parts are ranked

Attachments and conversation

Attached	Done in the app	Claude gets
STEP	CAD worker: extract, mesh, canonical pose, 6 renders; DINOv2. Not stored. Each solid → NEW-n; its vectors stay in the session for similar_parts	handle + size per solid, iso render (≤ 4)
Image	scaled to ≤ 1568 px, ≤ 5 MB (JPEG fallback)	image block; no image search
PDF, txt / md / csv	—	document block

- Each question resends the whole transcript, tool results included (cached), so a follow-up ("of those three…") can answer without tools. Citable ids carry over.
- An id repair keeps only the corrected answer; a refused or failed question is dropped. Past 200k chars of old tool results, those before the last 2 questions become a stub.
- New chat, an example, or a user switch starts a new conversation: nothing seen as admin stays.

Data

Four scattered tables (plm_items, erp_suppliers, doc_test_reports, doc_change_notices; 39% of notes Korean) are joined into one part card per part. 5,475 chunks: cards + one per note; each note embedded with a header (part, class, material, test, result) that is never cited.

Measured

- Permissions: 14 vs 33 parts for normal vs admin; a direct question about a P2 part leaked nothing.
- Ask 5–18 s per question, of which tools ~0.2 s: the latency is the LLM. A greeting: no tools, 7–17 s.
- Follow-up from history: 0 tools, 2.3 s. STEP attachment ~1 s (extract 0.2, mesh 0.05, render 0.2, embed 0.35 s); the sample bracket finds itself (BR-66254) at 1.0.
- ~$0.07 per question; one 60-question evaluation pass ≈ $4; concurrent runs via `ask_many`.

Retrieval

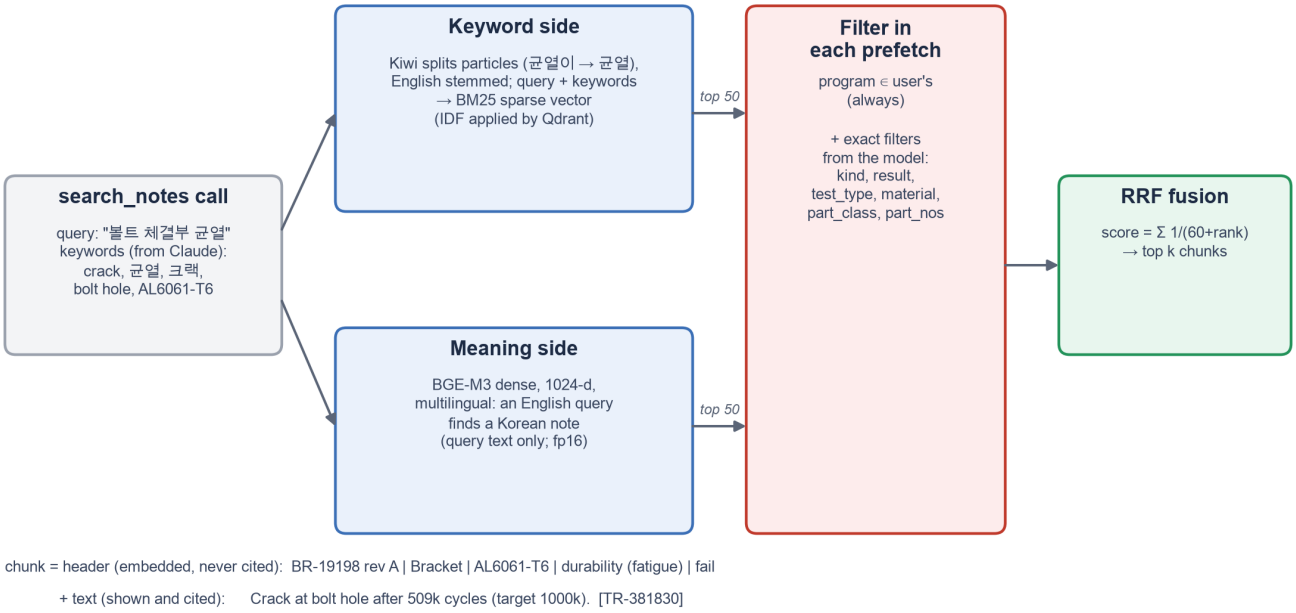


Figure 2. Text retrieval (search_notes, and the Search page without the LLM): keyword and meaning rankings, both restricted before ranking, merged by RRF.

Test	Result
크랙 난 브라켓 ('cracked bracket')	3 of top 5 said 'no crack'; fixed by result = fail filter
English query, BM25 / dense	BM25 matched wrong literal words; dense ignored the material
BM25 + bilingual expansion	4 of 4 correct, all from Korean notes

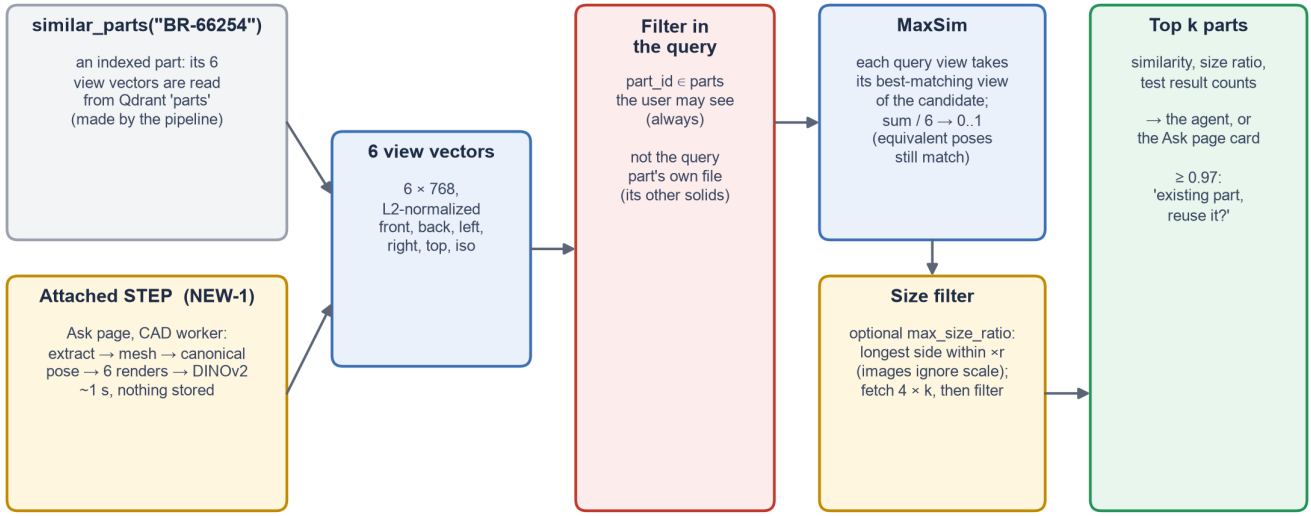


Figure 3. STEP retrieval (similar_parts, and the attachment card): one image signal, restricted before ranking, then filtered by size.